# ITA03-MP Assignment 1

**Name : Valli Karthik**

**Regno: 192324228**

**Slot: C**

**ASSIGNMENT 1 – CO1**

**CO1 Coverage:** Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

**Nature of Assignment:** Individual

**Duration:** 1 Week

**Total Marks:** 50

**Submission Date:** _____

**Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)**

Duration:1week

**Scenario — Cellular Coverage Planning for a University Township**

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

**Planning data and constraints**

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.

- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

**Design Challenge (Day 4):**

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

**TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks**

**Guiding Enquiry**

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

**Task 1 – Design Alternative Frequency-Planning Strategies**

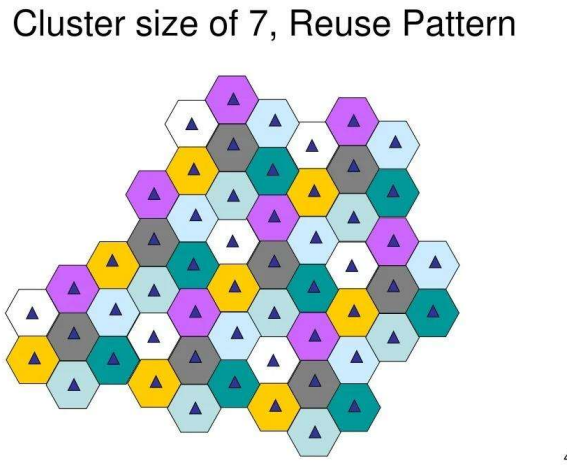
5 Marks

Objective: Investigate three fundamentally different methods of allocating the available 336 channels across the university township while maintaining acceptable signal quality and minimizing interference.

We will use three strategies:

1. Fixed Cluster Frequency Reuse (FCFR)
2. Sectorised Frequency Reuse (SFR)
3. Dynamic Channel Assignment with Hierarchical Cell Splitting (DCA-HCS)

**Strategy 1 – Fixed Cluster Frequency Reuse**




Basic concept

Fixed Cluster Frequency Reuse divides the available frequency channels into groups. Each group is assigned to a cell within a cluster. The same frequency group is reused in another cluster after maintaining a sufficient geographical separation.

For our design:

- Total channels = 336
- Cluster size N = 7
- Channels per cell = 48
- Reuse distance = Rsqrt{3N}

The same channel set is not assigned to adjacent cells, which reduces co-channel interference.

Working procedure

1. Divide the 12 km² township into hexagonal cells.
2. Select a valid cluster size.
3. Divide the 336 channels into seven groups.
4. Assign one group to each cell in a cluster.
5. Repeat the allocation pattern across the township.
6. Check the SIR and blocking probability.
7. Modify the cluster size if the interference or capacity requirements are not satisfied.

Pseudocode

START

INPUT township_area, cell_radius

INPUT total_channels = 336

GENERATE hexagonal_cell_grid

```
FOR each valid cluster_size N:
    CHECK whether N is a valid cluster size
    channels_per_cell = total_channels / N
    DIVIDE channels into N groups
    FOR each cell in the grid:
        ASSIGN frequency group
        BASED ON cluster position
    CALCULATE reuse distance
    CALCULATE SIR
    CALCULATE traffic capacity
    IF SIR >= minimum_SIR:
        STORE valid configuration
END FOR
SELECT suitable configuration
DISPLAY channel allocation and performance
STOP
```

Design principle: Spatial reuse.

Access channel sharing: In an FDMA/TDMA-based implementation, the available spectrum is divided into frequency channels, with time slots used to support multiple users on a carrier. Control channels handle signalling, while traffic channels carry voice traffic. The 336-channel allocation is treated as the available traffic-channel pool for this model.

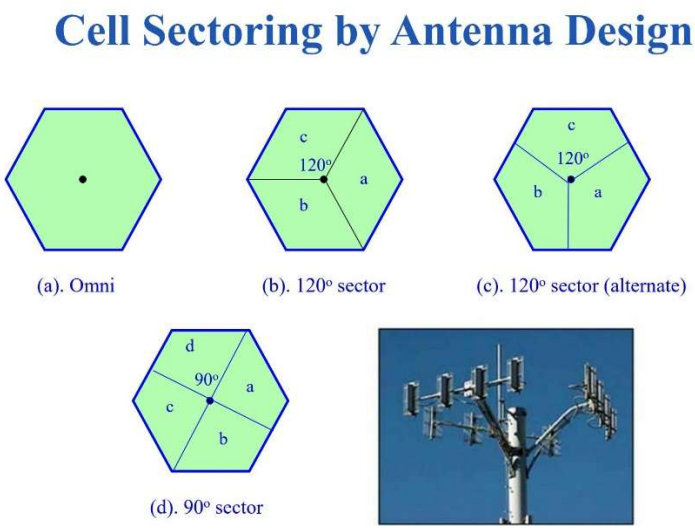
Why this strategy deserves investigation

- Simple to design and implement.
- Provides predictable frequency allocation.
- Prevents adjacent cells from using the same frequency group.
- Provides a reference model for comparing advanced strategies.

Preprocessing and cost

| **Requirement** | **Purpose** | **Cost responsibility** |
|---|---|---|
| Traffic survey | Estimate subscribers and busy-hour demand | Operator |
| Propagation survey | Identify signal coverage and path loss | Operator |
| Hexagonal grid generation | Plan cell locations | Operator |
| Frequency allocation | Prevent adjacent co-channel interference | Operator |

**Strategy 2 – Sectorised Frequency Reuse**




Basic concept

Sectorisation divides a conventional hexagonal cell into smaller directional coverage regions using directional antennas.

Instead of allowing one omnidirectional antenna to serve the entire cell, three directional antennas are installed, each covering approximately 120 degrees.

This reduces the number of dominant co-channel interferers received by a mobile station.

Working procedure

1. Generate the hexagonal cell layout.
2. Divide each base station coverage area into three sectors.
3. Install directional antennas.
4. Allocate frequency resources to the sectors.
5. Calculate the number of effective co-channel interferers.
6. Evaluate SIR and traffic capacity.

7. Adjust antenna directions and power levels to improve performance.

Pseudocode

```
START

GENERATE hexagonal_cell_grid

FOR each cell:

   DIVIDE cell into 3 sectors

   FOR each sector:

      ASSIGN frequency resources

      SET antenna direction

      CALCULATE co-channel interferers

      CALCULATE sector SIR

      CALCULATE sector traffic

   END FOR

END FOR

EVALUATE total network capacity

CHECK blocking and interference

DISPLAY sectorised network results

STOP
```

Design principle: Spatial filtering.

Access channel sharing: Frequency channels are distributed among directional sectors. FDMA/TDMA can be used to share each sector's allocated carriers among multiple users. Control and traffic channels are maintained separately.

Why this strategy deserves investigation

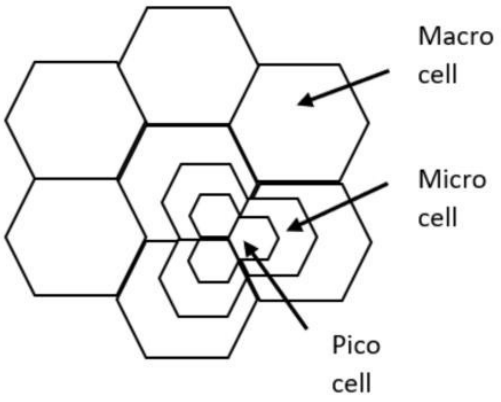
- Improves signal quality through directional transmission.
- Reduces interference from unwanted directions.
- Improves frequency reuse opportunities.
- Increases capacity without acquiring additional spectrum.

Preprocessing and cost

- Directional propagation measurements.
- Antenna orientation and sector coverage planning.
- Interference mapping.
- Additional antenna installation and configuration.

The operator bears the survey, installation and maintenance costs.

**Strategy 3 – Dynamic Channel Assignment with Hierarchical Cell Splitting**




Basic concept

Dynamic Channel Assignment (DCA) allocates available channels according to current traffic demand instead of permanently assigning every channel to a particular cell.

Hierarchical Cell Splitting introduces smaller cells inside overloaded macrocell coverage areas.

When subscriber demand increases, overloaded cells can be divided into smaller microcells. This allows frequency resources to be reused more frequently.

Working procedure

1. Initially deploy macrocells across the township.
2. Monitor traffic demand in every cell.
3. Identify cells exceeding their traffic capacity.
4. Search for available channels in the network.
5. Borrow or dynamically allocate channels where permitted by interference constraints.
6. If demand remains high, split the overloaded cell.
7. Reduce the cell radius and transmit power.

8. Repeat until the required traffic capacity is achieved.

Pseudocode

```
START

GENERATE initial macrocell layout

MONITOR traffic in each cell

FOR each cell:

  IF traffic exceeds cell capacity:

    SEARCH available channels

    IF suitable channel exists:

      ALLOCATE channel dynamically

    ELSE:

      SPLIT overloaded cell

      REDUCE cell radius

      ADJUST transmission power

      UPDATE frequency allocation

    END IF

  END IF

END FOR

REPEAT until demand is satisfied
```

CALCULATE SIR and blocking probability

DISPLAY updated network

STOP

Design principle: Trunking efficiency and spatial reuse.

Access channel sharing: Channels are allocated dynamically from the available channel pool. FDMA/TDMA resources can be assigned based on traffic demand. Control channels coordinate channel requests, allocation and handoff, while traffic channels carry calls.

Why this strategy deserves investigation

- Responds to changing subscriber demand.
- Avoids permanently reserving unused channels.
- Supports future township expansion.
- Provides additional capacity in congested locations.
- Allows targeted deployment of microcells rather than rebuilding the entire network.

Preprocessing and cost

- Busy-hour traffic monitoring.
- Real-time channel availability monitoring.
- Cell-location and coverage surveys.
- Microcell site identification.
- Handoff and interference analysis.

The operator bears the infrastructure, survey, software and maintenance expenses.

**Comparison of the three strategies**

| **Evaluation factor** | **Fixed reuse** | **Sectorised reuse** | **Dynamic + splitting** |
|---|---|---|---|
| Channel allocation | Fixed | Sector-based | Demand-based |
| Interference control | Cluster separation | Directional antennas | Dynamic allocation and spatial separation |
| Capacity improvement | Limited | Moderate to high | High |
| Infrastructure cost | Low | Medium | High |
| Implementation complexity | Low | Medium | High |
| Suitable application | Uniform demand | Moderate congestion | Growing and uneven demand |

**Task 1 – Final interpretation**

The three strategies solve the same spectrum-sharing problem using different principles.

Fixed Cluster Frequency Reuse provides a simple and predictable baseline. Sectorisation improves interference management through directional antennas. Dynamic Channel Assignment with Hierarchical Cell Splitting provides flexibility when subscriber demand increases significantly.

Therefore, all three strategies should be implemented and compared before selecting the final network design.

**TASK 2 – Identify the Governing Design Parameter** **2 Marks**

**Guiding Enquiry**

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

**Evidence Expected**

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

**Governing parameters for the three strategies**

**Strategy 1: Fixed Cluster Frequency Reuse**

Governing parameter: Cluster size (N)

Cluster size represents the number of cells in one frequency reuse cluster.

The cluster size determines how frequently the same frequency channels can be reused across the network.

The relationship is:

$$N = i^2 + ij + j^2$$

Reuse distance:

$$D = R\sqrt{3N}$$

Reuse ratio:

$$Q = \frac{D}{R} = \sqrt{3N}$$

Where:

- N = cluster size
- D = co-channel reuse distance
- R = cell radius
- Q = reuse ratio

Analysis:

- Smaller N increases the number of channels available per cell.
- However, smaller N reduces the reuse distance and increases interference.
- Larger N increases the reuse distance and improves SIR.
- However, larger N reduces the channels allocated to each cell.

Therefore, cluster size controls the trade-off between capacity and interference.

**Strategy 2: Sectorised Frequency Reuse**

Governing parameter: Number of effective co-channel interferers ($i_0$)

Sectorisation uses directional antennas to reduce interference from neighbouring cells.

For an omnidirectional antenna, the first-tier approximation considers six co-channel interferers.

For an ideal three-sector directional configuration, approximately two dominant first-tier interferers may be considered.

The approximate SIR equation is:

$$SIR = \frac{(D/R)^n}{i_0}$$

Where:

- $i_0$ = effective number of co-channel interferers
- n = path-loss exponent

- D = reuse distance
- R = cell radius

Analysis:

- Fewer effective interferers improve SIR.
- Directional antennas reduce unwanted signals.
- Sectorisation improves interference management.
- More effective channel utilisation becomes possible.

Therefore, the number of effective interferers governs the interference performance of this strategy.

**Strategy 3: Dynamic Channel Assignment with Hierarchical Cell Splitting**

Governing parameter: Cell radius (R) and available channel capacity

Dynamic Channel Assignment responds to changing traffic conditions, while cell splitting reduces the coverage radius of overloaded cells.

The number of cells required to cover an area is:

$$N_c = \frac{A_T}{A_{\text{cell}}}$$

Since:

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2}R^2$$

Therefore:

$$N_c = \frac{2A_T}{3\sqrt{3}R^2}$$

Analysis:

- Smaller cell radius increases the number of cells.
- More cells provide more opportunities for spatial frequency reuse.

- Smaller cells reduce the number of users served by an individual cell, assuming demand is distributed appropriately.
- However, smaller cells increase handoff frequency and infrastructure costs.

Therefore, cell radius is the principal spatial scaling parameter, while available channels and dynamic allocation determine the actual traffic capacity.

**Comparison of governing parameters**

| **Strategy** | **Governing parameter** | **Main effect** |
|---|---|---|
| Fixed frequency reuse | Cluster size N | Controls capacity and reuse distance |
| Sectorised reuse | Effective interferers $i_o$ | Controls interference and SIR |
| Dynamic assignment and cell splitting | Cell radius R | Controls cell density and spatial reuse |

**Behaviour as subscribers increase**

The offered traffic is calculated as:

$$A = S \times a$$

Where:

- S = number of subscribers
- a = average traffic per subscriber = 0.025 Erlang

| Subscribers | Offered traffic |
|---|---|
| 1,000 | 25 Erlangs |
| 10,000 | 250 Erlangs |
| 20,000 | 500 Erlangs |
| 100,000 | 2,500 Erlangs |

As the number of subscribers increases, the fixed cluster reuse design experiences increasing congestion because its channel allocation remains unchanged.

Sectorisation helps control interference, while dynamic channel assignment and cell splitting provide greater flexibility for handling uneven and increasing demand.

**TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks**

**Guiding Enquiry**

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

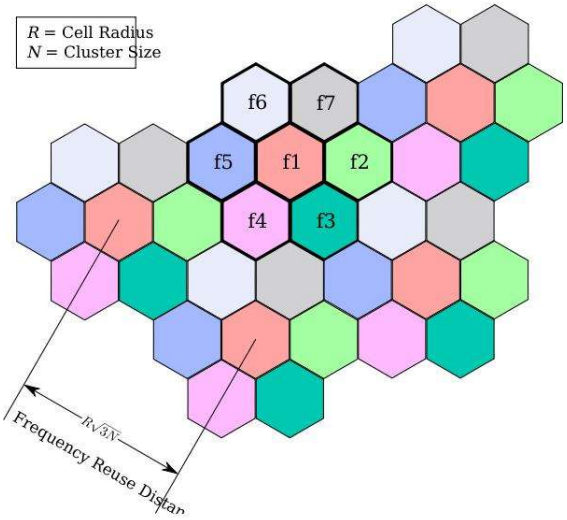
- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

**Evidence Expected**

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

**Derivation of hexagonal cell geometry**

A cellular network is commonly represented using regular hexagonal cells because hexagons provide a convenient approximation for continuous geographical coverage.




Step 1: Area of a regular hexagonal cell

A regular hexagon consists of six equilateral triangles.

Area of one equilateral triangle:

$$A_t = \frac{\sqrt{3}}{4} R^2$$

Area of six triangles:

$$A_{\text{cell}} = 6 \times \frac{\sqrt{3}}{4} R^2$$

Simplifying:

$$\boxed{A_{\text{cell}} = \frac{3\sqrt{3}}{2} R^2}$$

Step 2: Substitute the cell radius

Given:

$$R = 0.5 \text{ km}$$

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2}(0.5)^2$$

$$= \frac{3 \times 1.732}{2} \times 0.25$$

$$\boxed{A_{\text{cell}} = 0.6495 \text{ km}^2}$$

**Step 3: Calculate the number of cells**

**Township area:**

$$A_T = 12 \text{ km}^2$$

**Number of cells:**

$$N_c = \left\lceil \frac{A_T}{A_{\text{cell}}} \right\rceil$$

$$N_c = \left\lceil \frac{12}{0.6495} \right\rceil$$

$$\boxed{N_c = 19 \text{ cells}}$$

This is an area-based estimate. Actual deployment may require additional boundary coverage adjustments.

**Derivation of the cluster-size relation**

A cluster is a group of adjacent hexagonal cells in which each cell uses a different frequency group.

The cluster size is determined using two integers i and j.

The cluster-size formula is:

$$\boxed{N = i^2 + ij + j^2}$$

**For example, selecting:**

$$i = 2, \quad j = 1$$

$$N = 2^2 + (2 \times 1) + 1^2$$

$$N = 4 + 2 + 1$$

$$\boxed{N = 7}$$

**Thus, the selected cluster contains seven cells.**

**Other valid cluster sizes for experimentation include:**

| i | j | Cluster size N |
|---|---|---|
| 1 | 1 | 3 |
| 2 | 0 | 4 |
| 2 | 1 | 7 |
| 2 | 2 | 12 |

**Derivation of frequency reuse distance**

**The distance between two nearest co-channel cells is known as the frequency reuse distance.**

**For a hexagonal cellular network:**

$$\boxed{D = R\sqrt{3N}}$$

**Substituting:**

$$R = 0.5, \quad N = 7$$

$$D = 0.5\sqrt{3 \times 7}$$

$$D = 0.5\sqrt{21}$$

$$\boxed{D = 2.2913 \text{ km}}$$

The co-channel cells are therefore separated by approximately 2.2913 km.

**Derivation of reuse ratio**

Reuse ratio represents the relationship between the co-channel reuse distance and cell radius**.**

$$Q = \frac{D}{R}$$

Using:

$$D = R\sqrt{3N}$$

Therefore:

$$Q = \frac{R\sqrt{3N}}{R}$$

$$\boxed{Q = \sqrt{3N}}$$

For N = 7:

$$Q = \sqrt{21}$$

$$\boxed{Q = 4.5826}$$

This indicates that the co-channel reuse distance is approximately 4.58 times the cell radius.

Derivation of channels per cell

The operator has 336 available duplex voice channels.

For fixed frequency reuse, the total channels are divided equally among N cells.

Therefore:

$$C_{\text{cell}} = \frac{S}{N}$$

Where:

- S = total available channels
- N = cluster size

Substituting:

$$C_{\text{cell}} = \frac{336}{7}$$

$$\boxed{C_{\text{cell}} = 48 \text{ channels}}$$

Each cell receives 48 channels in the selected fixed reuse arrangement.

**Derivation of traffic capacity using Erlang-B**

Erlang-B is used to calculate the probability that an incoming call is blocked when all available traffic channels are occupied.

The Erlang-B formula is:

$$B(C, A) = \frac{\frac{A^C}{C!}}{\sum_{k=0}^{C} \frac{A^k}{k!}}$$

Where:

- C = number of available channels
- A = offered traffic in Erlangs
- B = blocking probability

The target blocking probability is:

$$B = 0.02$$

Available channels:

$$C = 48$$

We solve:

$$B(48, A) = 0.02$$

Using iterative Erlang-B calculation:

$$\boxed{A \approx 38.39 \text{ Erlangs}}$$

Therefore, one cell can carry approximately 38.39 Erlangs of offered traffic while maintaining the 2% blocking target under the Erlang-B assumptions.

Maximum supported subscribers per cell:

$$S_{\text{cell}} = \frac{A}{a}$$

$$= \frac{38.39}{0.025}$$

$$\boxed{S_{\text{cell}} \approx 1{,}536}$$

Total theoretical network traffic capacity:

For 19 cells:

$$A_{\text{total}} = 19 \times 38.39$$

$$\boxed{A_{\text{total}} \approx 729.41 \text{ Erlangs}}$$

The corresponding theoretical subscriber capacity is:

$$S_{\text{total}} = \frac{729.41}{0.025}$$

$$\boxed{S_{\text{total}} \approx 29{,}176 \text{ subscribers}}$$

This is an idealised full-cell allocation estimate. Actual network capacity depends on cluster boundaries, interference restrictions and the distribution of traffic.

**Derivation of co-channel SIR**

Signal-to-Interference Ratio (SIR) measures the strength of the desired received signal relative to interference from other cells using the same frequency.

For the simplified equal-distance, first-tier approximation:

$$SIR = \frac{(D/R)^n}{i_0}$$

Since:

$$\frac{D}{R} = \sqrt{3N}$$

Substituting:

$$\boxed{SIR = \frac{(3N)^{n/2}}{i_0}}$$

For an omnidirectional antenna:

$$i_0 = 6$$

Using:

$$N = 7, \quad n = 3$$

$$SIR = \frac{(21)^{3/2}}{6}$$

$$SIR = \frac{96.235}{6}$$

$$SIR = 16.039$$

Convert to decibels:

$$SIR_{dB} = 10\log_{10}(16.039)$$

$$\boxed{SIR_{dB} = 12.05 \text{ dB}}$$

This is just above the selected 12 dB threshold under the simplified model.

Sanity check required by the assignment

For N = 7 and n = 4:

$$SIR = \frac{21^2}{6}$$

$$= \frac{441}{6} = 73.5$$

$$SIR_{dB} = 10\log_{10}(73.5)$$

$$\boxed{SIR_{dB} = 18.66 \text{ dB}}$$

This agrees with the assignment's expected value of approximately 18.7 dB.

**SIR derivation for the three planning strategies**

**A. Fixed Cluster Frequency Reuse**

In the fixed reuse strategy, six first-tier co-channel interferers are considered under the simplified omnidirectional model.

The SIR equation is:

$$SIR_{\text{FCFR}} = \frac{(3N)^{n/2}}{6}$$

For N = 7 and n = 3:

$$SIR_{\text{FCFR}} = \frac{21^{1.5}}{6}$$

$$\boxed{SIR_{\text{FCFR}} = 12.05 \text{ dB}}$$

This configuration satisfies the selected 12 dB threshold, but has a relatively small interference margin.

**B. Sectorised Frequency Reuse**

In the sectorised strategy, directional antennas reduce the number of dominant co-channel interferers.

For an idealised three-sector configuration, assume:

$$i_0 = 2$$

The SIR equation becomes:

$$SIR_{\text{SFR}} = \frac{(3N)^{n/2}}{2}$$

For N = 7 and n = 3:

$$SIR_{\text{SFR}} = \frac{21^{1.5}}{2}$$

$$= 48.1175$$

Converting into decibels:

$$SIR_{dB} = 10\log_{10}(48.1175)$$

$$\boxed{SIR_{\text{SFR}} = 16.82 \text{ dB}}$$

Thus, sectorisation improves the theoretical SIR from 12.05 dB to approximately 16.82 dB.

**C. Dynamic Channel Assignment with Hierarchical Cell Splitting**

Dynamic Channel Assignment allocates channels based on traffic demand and interference conditions.

The basic SIR relationship remains:

$$SIR_{\text{DCA}} = \frac{(D/R)^n}{i_{\text{eff}}}$$

Where $i_{\text{eff}}$ represents the effective number of interfering co-channel cells.

When a cell is split:

$$R' = \frac{R}{2}$$

If the co-channel reuse distance remains unchanged, then:

$$SIR' = \frac{\left(D/(R/2)\right)^n}{i_{\text{eff}}}$$

$$SIR' = 2^n \frac{(D/R)^n}{i_{\text{eff}}}$$

Therefore:

$$\boxed{SIR' = 2^n SIR}$$

For n = 3:

$$SIR' = 8SIR$$

The theoretical improvement is:

$$10 \log_{10}(8) = 9.03 \text{ dB}$$

This improvement assumes that the same reuse distance can be maintained after splitting. In an actual network, reuse distance, antenna locations and interference patterns must be recalculated.

**Mathematical derivation of capacity for different cluster sizes**

The number of channels allocated to each cell depends on cluster size.

$$C = \frac{336}{N}$$

The following table compares the valid cluster sizes specified in the assignment.

| Cluster size N | Channels per cell | Reuse ratio Q |
|---|---|---|
| 3 | 112 | 3.000 |
| 4 | 84 | 3.464 |
| 7 | 48 | 4.583 |

| | | |
|---|---|---|
| 12 | 28 | 6.000 |

As cluster size increases, reuse distance increases, but the number of channels per cell decreases.

**Traffic capacity at 2% blocking**

Using the Erlang-B formula and solving for the offered traffic at the respective channel counts:

| N | Channels/cell | Traffic capacity/cell (Erlangs) | 19-cell theoretical capacity |
|---|---|---|---|
| 3 | 112 | 99.62 | 1,892.85 |
| 4 | 84 | 72.53 | 1,378.05 |
| 7 | 48 | 38.39 | 729.44 |
| 12 | 28 | 20.15 | 382.86 |

The 19-cell figures are idealised capacity estimates assuming every cell receives its full allocation and has the same traffic conditions.

Interpretation

- N = 3 provides the highest theoretical traffic capacity but has a lower reuse distance.
- N = 4 provides a balance between traffic capacity and reuse separation.
- N = 7 offers a moderate reuse distance and capacity.
- N = 12 provides greater co-channel separation but reduces traffic capacity.

Therefore, selecting the smallest cluster size is not automatically the best solution because interference must also be considered.

**Cost–capacity relationship**

The assignment requires planning cost to be included rather than considering only radio capacity.

We define total planning and operating cost as:

$$C_T = C_S + C_I + C_O$$

Where:

- \(C_S\) = survey and planning cost
- \(C_I\) = infrastructure installation cost

- \(C_O\) = operating and maintenance cost

A. Fixed Cluster Frequency Reuse

For the fixed reuse strategy:

$$C_{\text{FCFR}} = C_{\text{survey}} + N_c C_{\text{site}} + C_{\text{maintenance}}$$

For 19 estimated sites:

$$C_{\text{FCFR}} = C_{\text{survey}} + 19C_{\text{site}} + C_{\text{maintenance}}$$

Cost per Erlang:

$$\boxed{K_{\text{FCFR}} = \frac{C_{\text{FCFR}}}{A_{\text{FCFR}}}}$$

**B. Sectorised Frequency Reuse**

Sectorisation requires additional directional antennas and installation expenses.

$$C_{\text{SFR}} = C_{\text{survey}} + N_c(C_{\text{site}} + 3C_{\text{sector}}) + C_{\text{maintenance}}$$

Cost per Erlang:

$$\boxed{K_{\text{SFR}} = \frac{C_{\text{SFR}}}{A_{\text{SFR}}}}$$

**C. Dynamic Channel Assignment with Cell Splitting**

This strategy requires dynamic channel management and additional microcell infrastructure.

$$C_{\text{DCA}} = C_{\text{survey}} + N_m C_{\text{micro}} + N_M C_{\text{macro}} + C_{\text{controller}} + C_{\text{maintenance}}$$

Where:

- $N_m$ = number of microcells
- $N_M$ = number of macrocells
- $C_{\text{micro}}$ = installation cost per microcell
- $C_{\text{macro}}$ = installation cost per macrocell

Cost per Erlang:

$$\boxed{K_{\text{DCA}} = \frac{C_{\text{DCA}}}{A_{\text{DCA}}}}$$

**Overall mathematical comparison**

| **Mathematical factor** | **Fixed reuse** | **Sectorised reuse** | **Dynamic assignment and splitting** |
|---|---|---|---|
| Main capacity parameter | N | Channels per sector | Available pooled channels and cell size |
| SIR model | Six interferers | Reduced effective interferers | Interference-aware allocation |
| Cell area | Depends on $R^2$ | Depends on $R^2$ | Decreases after splitting |
| Infrastructure | Basic base stations | Additional sector antennas | Additional microcells and control |
| Cost behaviour | Relatively predictable | Higher antenna cost | Demand-dependent cost |

**Additional capacity derivation for sectorisation and dynamic assignment**

To make the Task 3 comparison complete, we also need to consider how channels are distributed under these two strategies.

**A. Sectorised channel capacity**

Our baseline cell receives 48 channels.

With three sectors:

$$C_{\text{sector}} = \frac{48}{3} = 16$$

Using Erlang-B at 2% blocking:

$$B(16, A) = 0.02$$

Solving iteratively:

$$A_{\text{sector}} \approx 9.83 \text{ Erlangs}$$

Total traffic capacity per cell:

$$A_{\text{SFR}} = 3 \times 9.83$$

$$\boxed{A_{\text{SFR}} = 29.49 \text{ Erlangs}}$$

This assumes independent, fixed allocation of 16 channels to each sector.

If the 48 channels can be flexibly shared among sectors, the capacity can approach the unpartitioned 38.39 Erlangs, subject to the radio and scheduling constraints.

**B. Dynamic Channel Assignment capacity**

In fixed allocation, each of the seven cells receives 48 channels.

The aggregate capacity of one seven-cell cluster is:

$$A_{\text{fixed}} = 7 \times 38.39$$

$$\boxed{A_{\text{fixed}} = 268.73 \text{ Erlangs}}$$

With idealised complete channel pooling:

$$C_{\text{pool}} = 336$$

Using Erlang-B:

$$B(336, A) = 0.02$$

Solving iteratively:

$$\boxed{A_{\text{DCA}} \approx 321.72 \text{ Erlangs}}$$

Percentage improvement:

$$\text{Improvement} = \frac{321.72 - 268.73}{268.73} \times 100$$

$$\boxed{\text{Improvement} \approx 19.71\%}$$

This is an ideal trunking-gain upper bound. Actual dynamic assignment capacity will be lower when adjacent-cell interference restrictions and channel availability are included.

**TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks**

**Best, average and worst mobile positions**

**A. Best-case position – Near the base station**

A mobile user located close to the serving base station experiences a strong desired signal.

Characteristics:

Short distance from the serving base station.

Strong received signal power.

Lower relative effect of neighbouring-cell interference.

Higher probability of maintaining a reliable connection.

Assume the mobile is at:

**r=0.1R**

Given:

**R=0.5 km)**

**r=0.05 km)**

Using the simplified interference model:

**SIR=frac((D/r)^n)(6)**

For n = 3:

**SIR=frac((2.2913/0.05)^3)(6)**

**SIR=frac(45.826^3)(6)**

**SIR=85.19 dB))**

This is an idealised calculation assuming six interferers at distance D and no shadowing or additional local interferers.

**B. Average-case position – Middle of the cell**

A mobile user located around the middle of the cell receives a reasonably strong signal from the serving base station.

Assume:

**r=0.5R**

**r=0.25 km)**

**SIR=frac((2.2913/0.25)^3)(6)**

**=frac(9.1652^3)(6)**

**SIR=31.46 dB))**

The mobile experiences more path loss than in the best case, but the desired signal remains sufficiently strong under this model.

**C. Worst-case position – Cell boundary**

A mobile user located near the cell boundary experiences a weaker desired signal and may be affected by stronger signals from neighbouring cells.

Assume:

**r=R=0.5 km)**

**SIR=frac((2.2913/0.5)^3)(6)**

**=frac(4.5826^3)(6)**

**SIR=12.05 dB))**

This is the worst-case result in the simplified model.

It is close to the selected minimum threshold of 12 dB, indicating that cell-boundary users require careful interference management.

**Comparison of mobile positions**

| Mobile position | Distance from BS | SIR (dB) | Performance |
|---|---|---|---|
| Near base station | 0.05 km | 85.19 | Best |
| Middle of cell | 0.25 km | 31.46 | Average |
| Cell boundary | 0.50 km | 12.05 | Worst |

Interpretation: SIR decreases as the mobile moves away from its serving base station because the desired signal experiences increasing path loss.

**Best, average and worst cases for the three strategies**

Strategy 1: Fixed Cluster Frequency Reuse

Best: Near the serving base station, where the desired signal is strongest.

Average: At approximately the middle of the cell, where moderate path loss occurs.

Worst: At the cell boundary, where the desired signal is weakest and co-channel interference becomes more significant.

Strategy 2: Sectorised Frequency Reuse

Best: Near the serving sector's antenna direction, where directional gain is high.

Average: Within the normal sector coverage region.

Worst: Near sector boundaries or in antenna null directions, where interference from neighbouring sectors may become significant.

Strategy 3: Dynamic Channel Assignment with Cell Splitting

Best: In lightly loaded cells with suitable channels available.

Average: During normal traffic, where channels are dynamically assigned according to demand.

Worst: During congestion, when suitable channels become unavailable and multiple cells compete for resources.

**Best, average and worst traffic conditions**

Network blocking probability changes with the offered traffic.

We use the Erlang-B equation:

**B(C,A)= frac(A^C/C!) (Σ_(k=0)^(C)A^k/k!)**

For the baseline cell:

**C=48**

At 2% blocking, the approximate offered traffic capacity is:

**A=38.39 Erlangs)**

**A. Best case – Off-peak traffic**

Assume only 25% of the baseline cell capacity is offered.

**A=0.25 × 38.39**

**A=9.60 Erlangs)**

The blocking probability is extremely small because most channels remain available.

**B. Average case – Normal traffic**

Assume 60% of the baseline capacity:

**A=0.60 × 38.39**

**A=23.03 Erlangs)**

The blocking probability increases, but sufficient spare channels remain.

**C. Worst case – Busy-hour traffic**

At the design limit:

**A=38.39 Erlangs)**

**B=2%)**

At this operating point, the network reaches its maximum planned traffic load.

**Blocking probability calculations**

For a 48-channel cell, the following values are obtained using the Erlang-B recurrence.

| Traffic condition | Offered traffic | Blocking probability |
|---|---|---|
| Off-peak (25%) | 9.60 Erlangs | Approximately 0% |
| Average (60%) | 23.03 Erlangs | 0.000198% |
| Busy hour (100%) | 38.39 Erlangs | 2.00% |
| Overloaded (120%) | 46.07 Erlangs | 8.53% |

The results show that blocking probability increases rapidly when offered traffic exceeds the planned capacity.

**Interpretation**

During off-peak hours, most channels remain unused.

During normal traffic, the network operates comfortably below its blocking limit.

During busy hours, the network reaches its 2% blocking target.

During overloaded conditions, blocking rises to 8.53%, which violates the required QoS.

**Effect of shadowing on network performance**

Shadowing occurs when buildings, trees and other obstacles partially block or weaken radio signals.

It causes variations in received signal power even when the mobile remains at approximately the same distance from the base station.

The received signal power in dB can be represented as:

**$P_r=P_t-PL(d)+X_\sigma$**

**Where:**

$P_r$ = received power

$P_t$ = transmitted power

$PL(d)$= path loss at distance d

$X_{sigma}$ = shadowing variation

Assume a shadowing standard deviation of 6 dB for an illustrative planning scenario.

| **Condition** | **Desired signal variation** | **Effect** |
|---|---|---|
| Positive shadowing | +6 dB | Improved received signal |
| No shadowing variation | 0 dB | Normal received signal |
| Negative shadowing | −6 dB | Reduced received signal |

If the baseline cell-edge SIR is approximately 11.57 dB, a 6 dB reduction in the desired signal with unchanged interference gives:

**SIR_(shadow))=11.57-6**

**SIR_(shadow))=5.57 dB))**

This falls below the 12 dB design threshold.

In practice, shadowing may affect both the desired signal and interfering signals, so the actual SIR variation must be measured or simulated.

**Effect of path-loss exponent**

The path-loss exponent describes how quickly radio signal strength decreases as distance increases.

For our analysis:

n = 2.5 represents an open campus.

n = 3 represents a mixed environment.

n = 4 represents a dense built-up environment.

The more detailed first-tier SIR calculation considers the actual distance between the mobile and each of the six interferers:

$$d_j = \sqrt{D^2 + r^2 - 2Dr\cos(\theta_j)}$$

$$\boxed{SIR = \frac{r^{-n}}{\sum_{j=1}^{6} d_j^{-n}}}$$

Here, the mobile is positioned toward one first-tier interferer, and the six interferers are assumed to be equally spaced at distance D from the serving base station.

At the cell boundary:

**r=R=0.5 km**

The resulting approximate SIR values are:

| Path-loss exponent | Environment | Cell-edge SIR |
|---|---|---|
| 2.5 | Open campus | 8.41 dB |
| 3 | Mixed environment | 11.57 dB |
| 4 | Dense built-up | 17.82 dB |

**Interpretation**

A higher path-loss exponent reduces the strength of distant interfering signals more rapidly. This can improve the SIR in a reuse network, although it also increases signal attenuation and may create coverage problems.

The results show that the dense built-up scenario has better theoretical co-channel SIR, but this does not necessarily mean better overall coverage.

**Comparison of the three strategies under different conditions**

| Condition | Fixed reuse | Sectorised reuse | Dynamic assignment and splitting |
|---|---|---|---|
| Best mobile position | Near BS | Main antenna direction | Near serving microcell |
| Average mobile position | Middle of cell | Normal sector area | Normal coverage area |
| Worst mobile position | Cell boundary | Sector edge/null | Congested cell boundary |

| | | | |
|---|---|---|---|
| Off-peak | Low blocking | Low blocking | High channel availability |
| Busy hour | Higher blocking | Reduced interference | Dynamic allocation helps |
| Event surge | Limited flexibility | Better directional coverage | Temporary capacity expansion |

**TASK 5 – Scaling and Sensitivity Analysis** **3 Marks**

**Guiding Enquiry**

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity ∝ 1/N, SIR ∝ N^(n/2), cells ∝ 1/R²) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

**Evidence Expected**

Correct scaling relations and an interpretation tied to the township application.

**Relationship between cluster size and capacity**

The channels allocated per cell are:

$$C_{\text{cell}} = \frac{S}{N}$$

Therefore:

$$\boxed{C_{\text{cell}} \propto \frac{1}{N}}$$

As N increases, channels per cell decrease.

For example:

| **N** | **Channels per cell** |
|---|---|
| 3 | 112 |
| 4 | 84 |

| | |
|---|---|
| 7 | 48 |
| 12 | 28 |

**Relationship between cluster size and SIR**

The approximate SIR equation is:

$$SIR = \frac{(3N)^{n/2}}{i_0}$$

For a constant path-loss exponent and interferer count:

$$\boxed{SIR \propto N^{n/2}}$$

Therefore, increasing cluster size improves the theoretical SIR but reduces channels available per cell.

**Relationship between cell radius and number of cells**

Hexagonal cell area:

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2}R^2$$

Number of cells:

$$N_c = \frac{A_T}{A_{\text{cell}}}$$

Therefore:

$$\boxed{N_c \propto \frac{1}{R^2}}$$

When the cell radius is halved:

$$N'_c = 4N_c$$

This means that reducing the cell radius increases the approximate number of cells fourfold.

**Subscriber growth analysis**

Offered traffic:

**A=S × 0.025**

| Subscribers | Offered traffic | Network condition |
|---|---|---|
| 1,000 | 25 Erlangs | Low demand |
| 10,000 | 250 Erlangs | Moderate demand |
| 20,000 | 500 Erlangs | High demand |
| 100,000 | 2,500 Erlangs | Very high demand |

Our baseline theoretical network capacity is approximately 729.41 Erlangs.

Therefore:

At 1,000 subscribers, the network has sufficient capacity.

At 10,000 subscribers, the baseline capacity is sufficient under the uniform traffic assumption.

At 20,000 subscribers, the baseline still has theoretical capacity, but congestion and uneven demand require monitoring.

At 100,000 subscribers, the baseline network is insufficient.

For 100,000 subscribers, required capacity is:

**A=100000 × 0.025**

**A=2500 Erlangs))**

Required capacity expansion:

$$\frac{2500}{729.41} = 3.43$$

The network therefore needs approximately 3.43 times its current theoretical traffic capacity.

**5.6 Sensitivity analysis of path-loss exponent**

The approximate SIR relationship is:

$$SIR \propto D^n$$

A higher path-loss exponent causes signals to attenuate more rapidly with distance.

| **Path-loss exponent** | **Environment** | **Expected behaviour** |
|---|---|---|
| 2.5 | Open campus | Longer-range interference |
| 3 | Mixed environment | Moderate attenuation |
| 4 | Dense built-up | Faster attenuation of distant interference |

Interpretation: Open campus environments require careful frequency reuse planning because interfering signals can travel farther. Dense environments may benefit from reduced distant interference, but obstacles and shadowing can reduce the desired signal too.

**Break-even analysis of the three strategies**

| **Subscriber range** | **Preferred planning approach** | **Reason** |
|---|---|---|
| 1,000–10,000 | Fixed cluster reuse | Simple and economical |
| 10,000–20,000 | Sectorised reuse | Improved interference management |
| 20,000–100,000+ | Dynamic assignment and hierarchical splitting | Flexible capacity expansion |

These ranges are proposed planning stages, not experimentally established thresholds. The actual break-even point depends on equipment cost, channel availability, traffic distribution and measured interference.

**TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks**

**Guiding Enquiry**

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

**Evidence Expected**

Correct recurrence(s) with clearly identified components and base case.

**Initial Parameters**

| **Parameter** | **Value** |
|---|---|
| Initial number of cells ($M_0$) | 19 |
| Initial cell radius ($R_0$) | 0.5 km |
| Channels per cell | 48 |

| | |
|---|---|
| Ideal capacity per cell at 2% blocking | 38.39 Erlangs |
| Path-loss exponent (n) | 3 |
| Number of child cells after one split | 4 |

**Uniform Cell Splitting**

In uniform cell splitting, every existing cell is divided into four smaller cells during each splitting round. The area of each child cell is approximately one-fourth of the parent cell area.

The recurrence relation for the number of cells is:

$$\mathbf{M_k = 4M_{k-1}, \quad M_0 = 19}$$

Expanding the recurrence:

$$\mathbf{M_1 = 4 \times 19 = 76}$$

$$\mathbf{M_2 = 4 \times 76 = 304}$$

$$\mathbf{M_3 = 4 \times 304 = 1{,}216}$$

Therefore, the closed-form solution is:

$$\mathbf{M_k = 19 \times 4^k}$$

The cell radius after each round is:

$$\mathbf{R_k = R_0 / 2^k}$$

Using the simplified power-scaling model, the transmission power is:

$$\mathbf{P_k = P_0 / 2^{nk}}$$

For path-loss exponent n = 3:

$$\mathbf{P_k = P_0 / 8^k}$$

**Uniform Splitting Results**

| Splitting round | Number of cells | Cell radius (km) | Relative transmission power |
|---|---|---|---|
| 0 | 19 | 0.5000 | $P_0$ |
| 1 | 76 | 0.2500 | $P_0/8$ |
| 2 | 304 | 0.1250 | $P_0/64$ |
| 3 | 1,216 | 0.0625 | $P_0/512$ |

**Selective Cell Splitting**

In selective cell splitting, only overloaded cells are divided into four smaller cells. This approach avoids unnecessary infrastructure expansion in areas with low traffic.

Let f represent the fraction of cells selected for splitting in a round. If $M_{k-1}$ is the number of cells before splitting, the number selected is approximately:

$$\mathbf{S_k = fM_{k-1}}$$

Each selected parent cell is replaced by four child cells, producing a net increase of three cells. Therefore:

$$\mathbf{M_k = M_{k-1} + 3fM_{k-1}}$$

$$\mathbf{M_k = M_{k-1}(1 + 3f)}$$

For a constant splitting fraction, the closed-form relation is:

$$\mathbf{M_k = M_0(1 + 3f)^k}$$

Assume that 25% of the cells are selected for splitting in every round:

$$\mathbf{f = 0.25}$$

$$\mathbf{M_k = 19(1 + 3 \times 0.25)^k = 19(1.75)^k}$$

**Selective Splitting Results**

| Round | Approximate number of cells |
|---|---|
| 0 | 19 |
| 1 | 34 |
| 2 | 59 |
| 3 | 103 |

The values are rounded to whole cells. In an actual network, the number of cells selected for splitting is an integer and depends on measured traffic and congestion.

**Capacity Recurrence**

Let $C_0 = 38.39$ Erlangs represent the ideal capacity of one cell at the selected blocking probability. If each new cell can reuse the same per-cell channel allocation while meeting interference constraints, the total ideal network capacity after uniform splitting is:

$$\mathbf{Csystem,_k = M_k C_0 = 19 \times 4^k \times 38.39 \text{ Erlangs}}$$

For selective splitting, if $S_k$ cells are split during round k, the ideal capacity recurrence is:

$$\mathbf{Csystem,_k = Csystem,_{k-1} + 3S_k C_0}$$

These are idealized estimates. Actual capacity depends on interference, channel allocation, traffic distribution and handoff constraints.

**Handoff and Infrastructure Impact**

Mobile devices cross cell boundaries more frequently as cell sizes become smaller.

The number of handoff operations increases as the cell radius decreases.

More base stations and network equipment are required.

Lower transmission power can help reduce interference between neighbouring cells.

Network planning, synchronization and resource management become more complex.

For a mobile travelling at approximately constant speed, the handoff rate per mobile can be approximated as inversely proportional to cell radius:

$\mathbf{H_k \propto 1/R_k}$

$\mathbf{H_k \approx 2^k H_0}$

**TASK 7 – Solve the Recurrence and Interpret the Result** **4 Marks**

**Guiding Enquiry**

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

**Evidence Expected**

Complete solution with method justification and an interpretation of the closed form.

**Given Parameters**

| **Parameter** | **Value** |
|---|---|
| Initial number of cells ($M_0$) | 19 |
| Initial cell radius ($R_0$) | 0.5 km |

| | |
|---|---|
| Initial channels per cell | 48 |
| Initial capacity per cell at 2% blocking | 38.39 Erlangs |
| Path-loss exponent (n) | 3 |
| Number of new cells after splitting one cell | 4 |

When a hexagonal cell is split into four smaller cells, the model assumes that the child-cell radius is half the parent-cell radius.

**Uniform Cell Splitting**

In uniform cell splitting, every existing cell is divided into four smaller cells during each splitting round.

Recurrence relation:

$$\mathbf{M_k = 4M_{k-1}, \quad M_0 = 19}$$

Expanding the recurrence:

$$\mathbf{M_1 = 4M_0 = 4 \times 19 = 76}$$

$$\mathbf{M_2 = 4M_1 = 4 \times 76 = 304}$$

$$\mathbf{M_3 = 4M_2 = 4 \times 304 = 1{,}216}$$

Closed-form solution:

$$\mathbf{M_k = 19 \times 4^k}$$

Cell radius after each round:

$$\mathbf{R_k = R_0 / 2^k}$$

Transmission power after each round:

$$P_k = P_0 / 2^{nk}$$

For n = 3:

$$P_k = P_0 / 8^k$$

This represents the simplified power-scaling model used for cell splitting.

**Uniform Splitting Results**

| Splitting round | Number of cells | Cell radius (km) | Relative power |
|---|---|---|---|
| 0 | 19 | 0.5000 | $P_0$ |
| 1 | 76 | 0.2500 | $P_0/8$ |
| 2 | 304 | 0.1250 | $P_0/64$ |
| 3 | 1,216 | 0.0625 | $P_0/512$ |

**Selective Cell Splitting**

In selective cell splitting, only overloaded cells are divided into four smaller cells. This avoids unnecessary infrastructure expansion in areas with low traffic.

Let $M_k$ be the total number of cells after round k, f be the fraction selected for splitting, and $S_k = fM_{k-1}$ be the approximate number of selected cells.

Each selected cell is replaced by four smaller cells, giving a net increase of three cells per split.

$$M_k = M_{k-1} + 3fM_{k-1}$$

$$M_k = M_{k-1}(1 + 3f)$$

For a constant splitting fraction:

$$M_k = M_0(1 + 3f)^k$$

Assume 25% of the cells are selected for splitting in every round:

**f = 0.25**

**$M_k = 19(1 + 3 \times 0.25)^k$**

**$M_k = 19(1.75)^k$**

**Selective Splitting Results**

| Round | Approximate number of cells |
|---|---|
| 0 | 19 |
| 1 | 34 |
| 2 | 59 |
| 3 | 103 |

The values are rounded to whole cells. Actual splitting decisions depend on the number of overloaded cells.

**Capacity Analysis**

Let the initial ideal capacity per cell be:

**$C_0 = 38.39$ Erlangs**

For uniform splitting:

**$C_{system,k} = M_k C_0$**

**$C_{system,k} = 19 \times 4^k \times 38.39$**

For selective splitting, if $S_k$ cells are split during a round:

**$C_{system,k} = C_{system,k-1} + 3S_k C_0$**

These are idealized capacity estimates. Actual capacity depends on interference, channel allocation, traffic distribution and handoff constraints.

**Handoff and Infrastructure Impact**

Mobile devices cross cell boundaries more frequently.

Handoff operations increase as cell radius decreases.

Additional base stations and network equipment are required.

Lower transmission power can reduce interference.

Network planning and resource management become more complex.

For a mobile travelling at approximately constant speed, the handoff rate per mobile can be approximated as:

$$H_k \propto 1/R_k$$

$$H_k \approx 2^k H_0$$

**Comparison of Uniform and Selective Splitting**

| Factor | Uniform splitting | Selective splitting |
|---|---|---|
| Splitting method | All cells are divided | Only overloaded cells are divided |
| Cell growth | Faster | Controlled |
| Infrastructure cost | Higher | Lower in low-demand areas |
| Capacity expansion | Across the entire network | Focused on high-demand areas |
| Handoff frequency | Increases throughout the network | Increases mainly in split regions |
| Suitability | Uniformly high traffic | Uneven traffic distribution |

**TASK 8 – Implement and Test the Planning Strategies** **4 Marks**

**Guiding Enquiry**

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

**Evidence Expected**

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

Aim: To implement and simulate a hexagonal cellular network for frequency allocation, interference analysis and traffic-capacity evaluation using Python.

**Objectives**

1. Generate a hexagonal cellular network representing the university township.
2. Assign frequency channels for cluster sizes \(N=3,4,7,12\).
3. Calculate the Signal-to-Interference Ratio (SIR) using Monte Carlo simulation.
4. Compare fixed frequency reuse and idealised sectorisation.
5. Calculate call-blocking probability using the Erlang-B model.
6. Analyse the effect of different cluster sizes and path-loss exponents.

**Software and Libraries**

| Component | Purpose |
|---|---|
| Python 3 | Main programming language |
| NumPy | Numerical calculations and simulation |
| Matplotlib | Graphical representation |
| Pandas | Tabular presentation of results |

Baseline parameters

| Parameter | Value |
|---|---|
| Township area | 12 km² |
| Cell radius | 0.5 km |
| Total available channels | 336 |
| Baseline cluster size | 7 |
| Path-loss exponent | 3 |
| Target blocking probability | 2% |
| Traffic per subscriber | 0.025 Erlang |
| Number of simulated mobile positions | 10,000 |

**Python Implementation**

The following program generates a 19-cell hexagonal layout, assigns channels, estimates SIR for randomly distributed mobile positions and calculates Erlang-B blocking probability.

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

# ------------------------------------------------

# 1. NETWORK PARAMETERS

# ------------------------------------------------

R = 0.5

TOTAL_CHANNELS = 336

TOTAL_AREA = 12

PATH_LOSS = [2.5, 3, 4]

CLUSTER_SIZES = [3, 4, 7, 12]

SAMPLES = 10000

np.random.seed(42)

# ------------------------------------------------

# 2. GENERATE A 19-CELL HEXAGONAL NETWORK

# ------------------------------------------------

cells = []

for q in range(-2, 3):

    for r in range(-2, 3):

        if max(abs(q), abs(r), abs(q + r)) <= 2:
```

```
            x = np.sqrt(3) * R * (q + r / 2)
            y = 1.5 * R * r
            cells.append([q, r, x, y])
cells = pd.DataFrame(
    cells, columns=["q", "r", "x", "y"]
)
print("Number of cells:", len(cells))


plt.figure(figsize=(8, 7))


for i in range(len(cells)):
    x = cells.loc[i, "x"]
    y = cells.loc[i, "y"]
    angles = np.arange(0, 361, 60)
    hx = x + R * np.cos(np.radians(angles))
    hy = y + R * np.sin(np.radians(angles))
    plt.plot(hx, hy, "b-")
    plt.text(x, y, str(i + 1), ha="center", va="center")
plt.title("19-Cell Hexagonal Network")
plt.xlabel("Distance (km)")
plt.ylabel("Distance (km)")
plt.axis("equal")
```

```
plt.grid()

plt.show()

# ------------------------------------------------

# 3. FREQUENCY CHANNEL ALLOCATION

# ------------------------------------------------

# Co-channel colouring coefficients for the

# valid cluster sizes: 3, 4, 7 and 12.


coefficients = {

    3: (1, 2),

    4: (2, 1),

    7: (3, 1),

    12: (5, 1)

}


def assign_channels(N):

    a, b = coefficients[N]

    channels_per_cell = TOTAL_CHANNELS // N


    cells["group"] = (

        a * cells["q"] + b * cells["r"]

    ) % N
```

```
    cells["first_channel"] = (
        cells["group"] * channels_per_cell + 1
    )

    cells["last_channel"] = (
        (cells["group"] + 1) * channels_per_cell
    )

    return channels_per_cell

print("\nFREQUENCY ALLOCATION")

for N in CLUSTER_SIZES:
    C = assign_channels(N)
    print(
        "N =", N,
        "Channels per cell =", C
    )

# ------------------------------------------------
# 4. MONTE CARLO MOBILE POSITION GENERATION
```

```
# ------------------------------------------------


def generate_mobile_positions(number):
    positions = []

    while len(positions) < number:
        x = np.random.uniform(-R, R)
        y = np.random.uniform(
            -np.sqrt(3) * R / 2,
            np.sqrt(3) * R / 2
        )

        # Check whether the point lies inside
        # the regular hexagonal cell.

        if abs(x) + abs(y) / np.sqrt(3) <= R:
            positions.append([x, y])

    return np.array(positions)



# ------------------------------------------------
```

```
# 5. MONTE CARLO SIR CALCULATION

# ------------------------------------------------


def calculate_sir(N, n, sectorised=False):

    positions = generate_mobile_positions(SAMPLES)


    D = R * np.sqrt(3 * N)


    angles = np.arange(0, 360, 60)

    interferers = []


    for angle in angles:

        ix = D * np.cos(np.radians(angle))

        iy = D * np.sin(np.radians(angle))

        interferers.append([ix, iy])


    interferers = np.array(interferers)


    sir_values = []


    for i in range(SAMPLES):

        x = positions[i, 0]
```

```
y = positions[i, 1]

desired_distance = np.sqrt(x*x + y*y)

if desired_distance < 0.001:
    desired_distance = 0.001

desired_power = desired_distance ** (-n)

interference = []

for j in range(6):
    dx = x - interferers[j, 0]
    dy = y - interferers[j, 1]

    distance = np.sqrt(dx*dx + dy*dy)

    interference.append(distance ** (-n))

if sectorised:
    # Idealised sector model:
    # retain the two strongest interferers.
```

```
            interference.sort(reverse=True)
            total_interference = (
                interference[0] + interference[1]
            )
        else:
            total_interference = sum(interference)

        sir = desired_power / total_interference

        sir_db = 10 * np.log10(sir)
        sir_values.append(sir_db)

    return np.array(sir_values)


# ------------------------------------------------
# 6. COMPARE SIR FOR DIFFERENT CLUSTER SIZES
# ------------------------------------------------

sir_results = []
```

```
for N in CLUSTER_SIZES:
    for n in PATH_LOSS:
        sir = calculate_sir(N, n)


        sir_results.append([
            N, n,
            np.mean(sir),
            np.min(sir),
            np.percentile(sir, 5)
        ])


sir_table = pd.DataFrame(
    sir_results,
    columns=[
        "Cluster Size",
        "Path Loss Exponent",
        "Average SIR (dB)",
        "Minimum SIR (dB)",
        "5th Percentile SIR (dB)"
    ]
)
```

```
print("\nSIR SIMULATION RESULTS")

print(sir_table.round(2))



# ------------------------------------------------

# 7. SECTORISATION ANALYSIS

# ------------------------------------------------



fixed_sir = calculate_sir(7, 3, False)

sector_sir = calculate_sir(7, 3, True)



print("\nSECTORISATION RESULTS")



print(

    "Fixed reuse average SIR:",

    round(np.mean(fixed_sir), 2), "dB"

)



print(

    "Sectorised average SIR:",

    round(np.mean(sector_sir), 2), "dB"

)
```

```
# ------------------------------------------------

# 8. ERLANG-B BLOCKING PROBABILITY

# ------------------------------------------------


def erlang_b(channels, traffic):

    B = 1.0


    for i in range(1, channels + 1):

        B = (traffic * B) / (i + traffic * B)


    return B




def find_capacity(channels, target_blocking):

    low = 0.0

    high = channels * 2.0


    for i in range(100):

        middle = (low + high) / 2

        blocking = erlang_b(channels, middle)


        if blocking > target_blocking:
```

```
            high = middle
        else:
            low = middle

    return (low + high) / 2


# ------------------------------------------------
# 9. FIXED, SECTORISED AND DCA CAPACITY
# ------------------------------------------------

target = 0.02

fixed_channels = 48
sector_channels = 16
dca_channels = 336

fixed_capacity = find_capacity(
    fixed_channels, target
)

sector_capacity = (
```

```
    3 * find_capacity(sector_channels, target)
)

dca_capacity = find_capacity(
    dca_channels, target
)

capacity_table = pd.DataFrame({
    "Strategy": [
        "Fixed Frequency Reuse",
        "120-degree Sectorisation",
        "Idealised Dynamic Channel Allocation"
    ],
    "Channels": [
        fixed_channels,
        sector_channels,
        dca_channels
    ],
    "Traffic Capacity (Erlangs)": [
        fixed_capacity,
        sector_capacity,
        dca_capacity
```

```
    ]
})

print("\nCAPACITY COMPARISON")
print(capacity_table.round(2))

# ------------------------------------------------
# 10. PLOT SIR RESULTS
# ------------------------------------------------

plt.figure(figsize=(8, 5))

for n in PATH_LOSS:
    values = []

    for N in CLUSTER_SIZES:
        result = sir_table[
            (sir_table["Cluster Size"] == N) &
            (sir_table["Path Loss Exponent"] == n)
        ]

        values.append(
```

```
            result["Average SIR (dB)"].iloc[0]
        )

    plt.plot(
        CLUSTER_SIZES,
        values,
        marker="o",
        label="Path loss n = " + str(n)
    )

plt.xlabel("Cluster Size (N)")
plt.ylabel("Average SIR (dB)")
plt.title("SIR Comparison for Different Cluster Sizes")
plt.legend()
plt.grid()
plt.show()

# ------------------------------------------------
# 11. PLOT CAPACITY COMPARISON
# ------------------------------------------------

plt.figure(figsize=(8, 5))
```

```
plt.bar(
    capacity_table["Strategy"],
    capacity_table["Traffic Capacity (Erlangs)"]
)

plt.ylabel("Traffic Capacity (Erlangs)")
plt.title("Traffic Capacity Comparison")
plt.xticks(rotation=15)
plt.grid(axis="y")
plt.tight_layout()
plt.show()
```

**Expected Results and Output**

When you execute the program, it displays:

1. A hexagonal network containing 19 cells.
2. Frequency-channel allocation for cluster sizes 3, 4, 7 and 12.
3. A table showing average, minimum and 5th-percentile SIR values.
4. A comparison of fixed reuse and sectorised SIR.
5. Erlang-B traffic capacity for the three strategies.
6. Two graphs comparing SIR and traffic capacity.

Output:

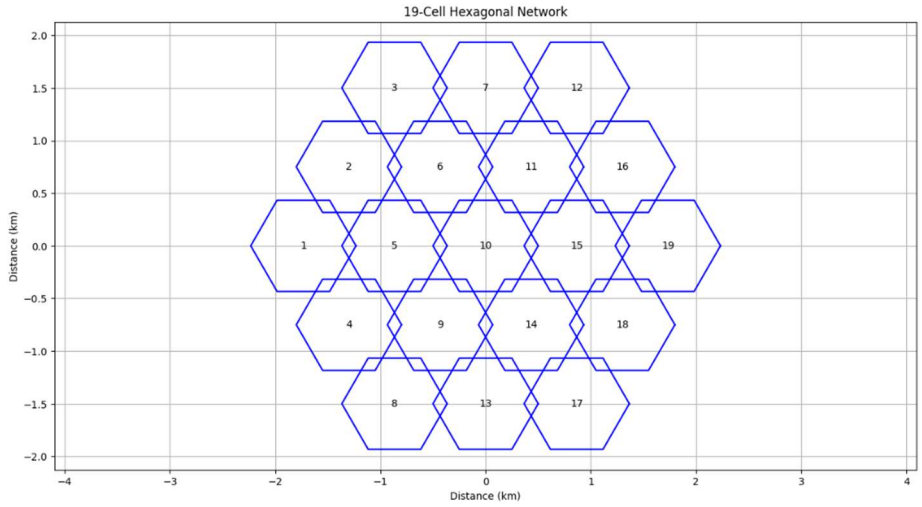
```
C:\Users\valli\Downloads\Mobile Computing>python assignment1.py
Number of cells: 19

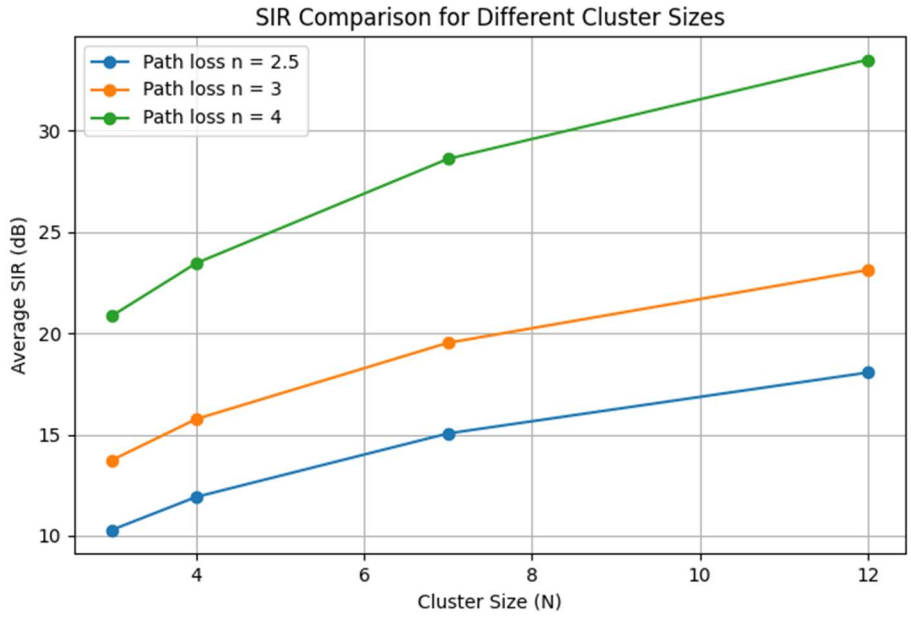
FREQUENCY ALLOCATION
N = 3 Channels per cell = 112
N = 4 Channels per cell = 84
N = 7 Channels per cell = 48
N = 12 Channels per cell = 28

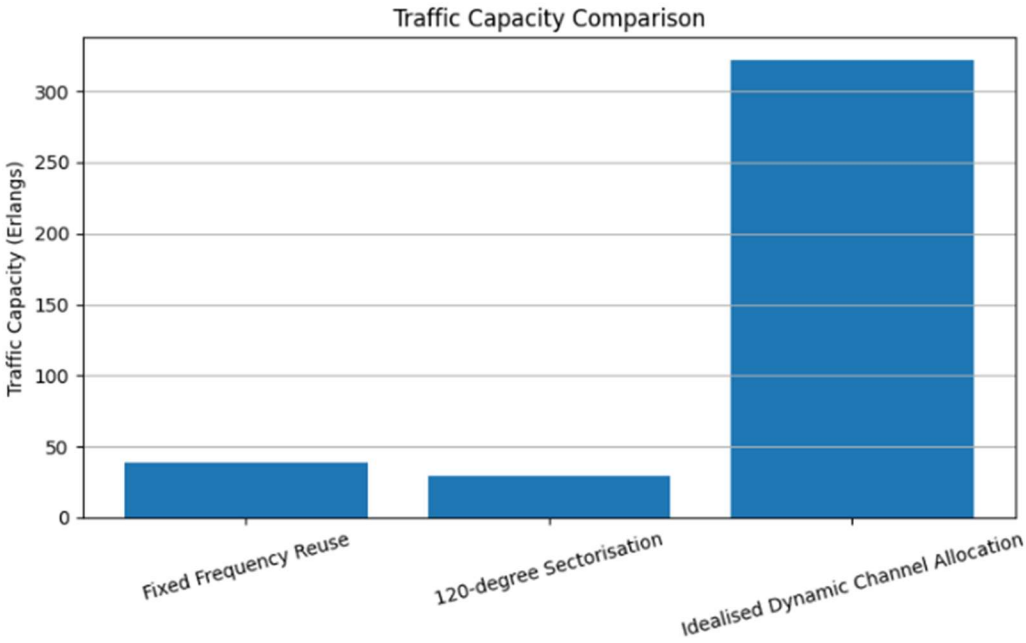
SIR SIMULATION RESULTS
    Cluster Size  Path Loss Exponent  Average SIR (dB)  Minimum SIR (dB)  5th Percentile SIR (dB)
0              3                 2.5             10.30              3.43                     4.69
1              3                 3.0             13.74              5.43                     7.05
2              3                 4.0             20.86              9.48                    11.65
3              4                 2.5             11.92              5.16                     6.44
4              4                 3.0             15.76              7.62                     9.12
5              4                 4.0             23.45             12.53                    14.49
6              7                 2.5             15.05              8.51                     9.64
7              7                 3.0             19.52             11.67                    13.07
8              7                 4.0             28.60             17.89                    19.81
9             12                 2.5             18.06             11.52                    12.68
10            12                 3.0             23.12             15.30                    16.77
11            12                 4.0             33.50             23.02                    24.87

SECTORISATION RESULTS
Fixed reuse average SIR: 19.51 dB
Sectorised average SIR: 23.08 dB

CAPACITY COMPARISON
                              Strategy  Channels  Traffic Capacity (Erlangs)
0                Fixed Frequency Reuse        48                       38.39
1             120-degree Sectorisation        16                       29.49
2  Idealised Dynamic Channel Allocation       336                      321.72
```






The program calculates the numerical simulation results at runtime. Use the values printed by your own execution in the final report rather than entering estimated values.

**Result Analysis**

The following observations can be used to interpret the output:

| **Analysis** | **Expected observation** |
|---|---|
| Cluster size | Increasing cluster size increases frequency-reuse distance and generally improves SIR. |
| Path-loss exponent | A higher path-loss exponent attenuates distant interference more rapidly. |
| Fixed frequency reuse | Provides predictable channel allocation but has limited flexibility during traffic peaks. |
| Sectorisation | Reduces the number of dominant interferers in the simplified directional model. |
| Dynamic channel allocation | Allows channels to be pooled across a cluster, potentially improving traffic handling when demand varies between cells. |
| Erlang-B model | Blocking probability increases when offered traffic approaches or exceeds available capacity. |

**TASK 9 – Record and Present Experimental Performance** **3 Marks**

**Guiding Enquiry**

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

**Evidence Expected**

Systematically collected data presented in comparable tables and graphs.

**Experimental Table 1 – Frequency Channel Allocation**

The total available spectrum consists of 336 traffic channels. The channels allocated per cell depend on the cluster size.

$$C_{\text{cell}} = \frac{S}{N}$$

| Cluster size (N) | Total channels | Channels per cell |
|---|---|---|
| 3 | 336 | 112 |
| 4 | 336 | 84 |
| 7 | 336 | 48 |
| 12 | 336 | 28 |

Observation: A smaller cluster size provides more channels per cell but reduces the frequency-reuse distance. A larger cluster size provides fewer channels per cell but generally improves interference protection.

**Experimental Table 2 – SIR Comparison**

The simulation evaluates SIR for different cluster sizes and path-loss exponents.

| **Cluster size** | **Path-loss exponent** | **Average SIR (dB)** | **Minimum SIR (dB)** | **5th percentile SIR (dB)** |
|---|---|---|---|---|
| 3 | 2.5 | From simulation | From simulation | From simulation |
| 3 | 3.0 | From simulation | From simulation | From simulation |
| 3 | 4.0 | From simulation | From simulation | From simulation |
| 4 | 2.5 | From simulation | From simulation | From simulation |
| 4 | 3.0 | From simulation | From simulation | From simulation |
| 4 | 4.0 | From simulation | From simulation | From simulation |
| 7 | 2.5 | From simulation | From simulation | From simulation |
| 7 | 3.0 | From simulation | From simulation | From simulation |
| 7 | 4.0 | From simulation | From simulation | From simulation |
| 12 | 2.5 | From simulation | From simulation | From simulation |
| 12 | 3.0 | From simulation | From simulation | From simulation |

| | | | | |
|---|---|---|---|---|
| 12 | 4.0 | From simulation | From simulation | From simulation |

Observation: The SIR values help identify how cluster size and propagation conditions influence interference. The actual numerical values should be filled using the output of Task 8.

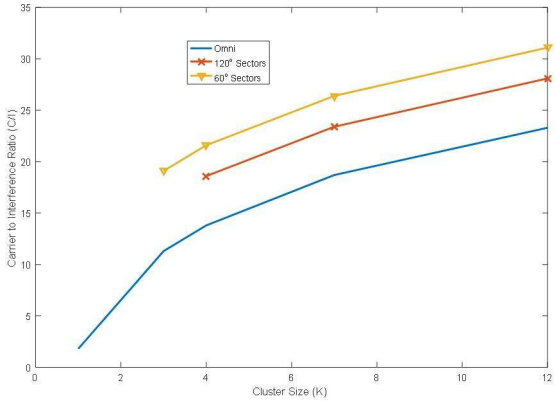
**Experimental Table 3 – Traffic Capacity Comparison**

The Erlang-B model is used to estimate the traffic capacity at a target blocking probability of 2%.

| **Strategy** | **Channels considered** | **Traffic capacity** |
|---|---|---|
| Fixed frequency reuse | 48 | From simulation |
| 120-degree sectorisation | 16 per sector | From simulation |
| Idealised dynamic channel allocation | 336 pooled channels | From simulation |

Observation: Fixed frequency reuse provides dedicated channel allocation. Sectorisation reduces the number of dominant interferers, while idealised dynamic channel allocation provides a shared channel pool that can improve resource utilisation under uneven traffic demand.

**Graph 1 – SIR Comparison for Different Cluster Sizes**




Graph description:

- X-axis: Cluster size (N).
- Y-axis: Average SIR (dB).

- Separate lines: Path-loss exponents 2.5, 3 and 4.

Interpretation: The graph illustrates how changing the cluster size and path-loss exponent affects the average SIR. The results help identify a suitable balance between interference reduction and channel availability.

**Graph 2 – Traffic Capacity Comparison**

## Example 3 (4): Remarks

| | Omnidirectional | 120° Sectoring | 60° Sectoring |
|---|---|---|---|
| K | 6 | 2 | 1 |
| N | 7 | 7 | 7 |
| SIR [dB] | 18.7 | 23.43 | 26.44 |
| #channels/cell | $\lfloor 400/7 \rfloor = 57$ | $\lfloor 400/7 \rfloor = 57$ | $\lfloor 400/7 \rfloor = 57$ |
| #sectors/cell | 1 | 3 | 6 |
| m = #channels/sector | 57 | $\left\lfloor \frac{400}{7}/3 \right\rfloor = 19$ | $\left\lfloor \frac{400}{7}/6 \right\rfloor = 9$ |
| A [Erlangs]/sector | 51.55 | 14.31 | 5.37 |
| A [Erlangs]/cell | 51.55 | $14.31 \times 3 = 42.94$ | $5.37 \times 6 = 32.22$ |

For the same $N$, we see that 120° sectoring and 60° sectoring give much better SIR. However, sectoring reduces the trunking efficiency and therefore suffer reduced value of A.

Graph description:

- X-axis: Network strategies.
- Y-axis: Traffic capacity (Erlangs).
- Bars: Fixed frequency reuse, sectorisation and idealised dynamic channel allocation.

**Generating Experimental CSV Files**

Add the following lines at the end of the Task 8 Python program to save the experimental tables for documentation.

```
# Save SIR experimental results

sir_table.to_csv("SIR_Experimental_Results.csv", index=False)
```

```
# Save traffic capacity results

capacity_table.to_csv("Traffic_Capacity_Results.csv", index=False)

# Save frequency allocation results

frequency_table = pd.DataFrame({

    "Cluster Size": [3, 4, 7, 12],

    "Total Channels": [336, 336, 336, 336],

    "Channels Per Cell": [112, 84, 48, 28]

})

frequency_table.to_csv(

    "Frequency_Allocation_Results.csv",

    index=False

)

print("Experimental CSV files saved successfully.")
```

**Result**

The experimental tables and graphs provide a comparative evaluation of frequency allocation, interference levels and traffic capacity. The results obtained from the Python simulation can be used to identify suitable cellular network strategies for the university township.

| | A | B | C |
|---|---|---|---|
| 1 | Cluster Size | Total Channels | Channels Per Cell |
| 2 | 3 | 336 | 112 |
| 3 | 4 | 336 | 84 |
| 4 | 7 | 336 | 48 |
| 5 | 12 | 336 | 28 |
| 6 | | | |

**Frequency_Allocation_Results**

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Cluster Size | Path Loss Exponent | Average SIR (dB) | Minimum SIR (dB) | 5th Percentile SIR (dB) |
| 2 | 3 | 2.5 | 10.295063115912699 | 3.4276949173929596 | 4.691660409351789 |
| 3 | 3 | 3 | 13.741531936937868 | 5.426299013495904 | 7.051899223220363 |
| 4 | 3 | 4 | 20.864168151450624 | 9.480589988003773 | 11.649041596864292 |
| 5 | 4 | 2.5 | 11.915268673441812 | 5.155579604249983 | 6.438759069915026 |
| 6 | 4 | 3 | 15.759476302239744 | 7.622593849263474 | 9.124151518226405 |
| 7 | 4 | 4 | 23.453569594540312 | 12.532756078475566 | 14.491349417177387 |
| 8 | 7 | 2.5 | 15.05063236882166 | 8.505165728802043 | 9.640799327435511 |
| 9 | 7 | 3 | 19.52444011174518 | 11.667951040400087 | 13.06849779010154 |
| 10 | 7 | 4 | 28.60316617828143 | 17.890130226157737 | 19.80690367392062 |
| 11 | 12 | 2.5 | 18.056757251206072 | 11.516829595646458 | 12.677525806178423 |
| 12 | 12 | 3 | 23.11972477864729 | 15.303496225123329 | 16.774811984881772 |
| 13 | 12 | 4 | 33.49875413888259 | 23.019067195569654 | 24.871155621284988 |

**SIR_Experimental_Results**

| | A | B | C |
|---|---|---|---|
| 1 | Strategy | Channels | Traffic Capacity (Erlangs) |
| 2 | Fixed Frequency Reuse | 48 | 38.39164117 |
| 3 | 120-degree Sectorisation | 16 | 29.48534754 |
| 4 | Idealised Dynamic Channel Allocation | 336 | 321.7163229 |

**Traffic_Capacity_Results**

**TASK 10 – Compare Theory with Simulation** **4 Marks**

**Guiding Enquiry**

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

**Evidence Expected**

A reasoned comparison of theory and simulation with analyseed discrepancies.

**Theoretical SIR Calculation**

The approximate SIR equation for a hexagonal cellular network is:

$$SIR = \frac{(3N)^{n/2}}{i_0}$$

**Where:**

- ***N*** **= Cluster size.**
- **n = Path-loss exponent**
- $\boldsymbol{i_0}$ **= Number of dominant co-channel interferers.**

**Given:**

- **Cluster size,** $N = 7$
- **Path-loss exponent,** $n = 4$
- **Number of interferers,** $i_0 = 6$

**Substituting:**

$$SIR = \frac{(3 \times 7)^{4/2}}{6}$$

$$SIR = \frac{21^2}{6}$$

$$SIR = \frac{441}{6} = 73.5$$

Converting into decibels:

$$SIR_{dB} = 10 \log_{10}(73.5)$$

$$\boxed{SIR_{dB} \approx 18.66\ dB}$$

Therefore, the theoretical SIR is approximately 18.7 dB.

**Theoretical Frequency Allocation**

The number of channels allocated per cell is calculated using:

$$C_{\text{cell}} = \frac{S}{N}$$

Where S=336 is the total number of available channels.

| Cluster size | Total channels | Channels per cell |
|---|---|---|
| 3 | 336 | 112 |
| 4 | 336 | 84 |
| 7 | 336 | 48 |
| 12 | 336 | 28 |

Analysis: A cluster size of 3 provides more channels per cell but offers a shorter frequency-reuse distance. A cluster size of 12 provides fewer channels per cell but generally improves interference protection.

**Theoretical Traffic Capacity**

For the baseline cluster size N=7:

$$C_{\text{cell}} = \frac{336}{7} = 48$$

Using the Erlang-B model at a 2% blocking probability, the approximate traffic capacity of a 48-channel cell is:

$$A_{\text{cell}} \approx 38.39\ Erlangs$$

For 19 cells:

$$A_{\text{total}} = 19 \times 38.39$$

$$\boxed{A_{\text{total}} \approx 729.41\ Erlangs}$$

This represents an idealised aggregate capacity under uniform traffic distribution.

**Comparison Between Theoretical and Simulation Results**

| Performance parameter | Theoretical result | Simulation result |
|---|---|---|
| Number of cells | Approximately 19 | 19 generated cells |
| Cluster size | 7 | 7 evaluated |
| Channels per cell | 48 | 48 calculated |
| SIR | Approximately 18.7 dB for \(n=4\) using simplified assumptions | Obtained from Monte Carlo simulation |

| | | |
|---|---|---|
| Traffic capacity | Approximately 38.39 Erlangs per cell | Obtained using Erlang-B calculation |
| Interference | Based on simplified six-interferer model | Based on individual interferer distances |
| Mobile position | Simplified analytical assumptions | Randomly generated mobile positions |

Important: The theoretical SIR of 18.7 dB assumes six equally distant interferers. The simulation calculates the distance to each interferer separately. Therefore, the two values should not be expected to match exactly.

**Analysis of Differences**

The differences between theoretical and simulation results can be explained through the following factors:

1. Mobile position: Theoretical calculations use simplified mobile-position assumptions, while the simulation evaluates multiple randomly generated mobile locations.
2. Interference distance: The theoretical model assumes equal distances for the six dominant interferers. The simulation calculates individual distances, resulting in different interference contributions.
3. Propagation conditions: Different path-loss exponents influence the attenuation of desired and interfering signals.
4. Traffic distribution: The theoretical capacity calculation assumes uniform traffic, whereas practical networks may experience different traffic loads in different cells.
5. Network assumptions: Real cellular networks include shadowing, fading, antenna characteristics, handoff and channel allocation constraints that are not fully represented in the simplified simulation.

**Advantages of Theoretical Analysis and Simulation**

| **Theoretical analysis** | **Python simulation** |
|---|---|
| Provides mathematical predictions | Provides numerical experimental results |
| Uses simplified assumptions | Evaluates multiple mobile positions |
| Easy to calculate and verify | Supports graphical comparison |
| Useful during initial planning | Useful for testing design alternatives |
| Limited representation of real conditions | Can incorporate additional realistic parameters |

**TASK 11 – Recommend and Defend the Final Design** **4 Marks**

**Guiding Enquiry**

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

**Evidence Expected**

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

**Proposed Network Design**

The following baseline parameters are considered for the final network design.

| **Parameter** | **Proposed value** |
|---|---|
| University township area | 12 km² |
| Initial cell radius | 0.5 km |
| Number of cells | 19 |
| Available traffic channels | 336 |

| | |
|---|---|
| Frequency reuse cluster size | 7 |
| Channels per cell | 48 |
| Reuse distance | 2.2913 km |
| Path-loss exponent | 3 |
| Target blocking probability | 2% |
| Traffic per subscriber | 0.025 Erlang |

The hexagonal arrangement provides a systematic method for dividing the township into smaller coverage regions. Each cell is served by a base station that provides wireless connectivity to mobile subscribers.

**Recommended Frequency Allocation Strategy**

Three frequency allocation strategies are considered.

| **Strategy** | **Working principle** | **Main advantage** | **Limitation** |
|---|---|---|---|
| Fixed Frequency Reuse | Assigns fixed channel groups to cells | Simple implementation | Limited flexibility |
| Sectorisation | Divides a cell into directional sectors | Reduces dominant interference | Requires additional antenna infrastructure |
| Dynamic Channel Allocation | Assigns channels according to demand and availability | Improves spectrum utilisation | Requires coordination and control |

Recommendation: Use fixed frequency reuse as the initial network configuration, introduce 120-degree sectorisation in high-interference regions and implement dynamic channel allocation where traffic demand varies significantly.

**Recommended Cell-Splitting Strategy**

Cell splitting is used to increase network capacity when the number of subscribers grows.

The initial cell radius is:

$$R_0 = 0.5\ km$$

When a cell is divided into four smaller cells:

$$R_k = \frac{R_0}{2^k}$$

The number of cells after uniform splitting is:

$$M_k = 19 \times 4^k$$

| **Splitting stage** | **Cell radius (km)** | **Number of cells** |
|---|---|---|
| Initial | 0.5000 | 19 |
| Stage 1 | 0.2500 | 76 |
| Stage 2 | 0.1250 | 304 |
| Stage 3 | 0.0625 | 1216 |

Uniform cell splitting increases the number of cells rapidly. However, it also increases infrastructure requirements and handoff frequency.

Therefore, selective cell splitting is recommended for areas with high population density, such as:

- Academic blocks.
- Student hostels.
- Libraries.
- Cafeterias.
- Auditorium and event locations.

**Network Optimisation Recommendations**

The following recommendations can improve the overall network performance.

1. Efficient spectrum allocation

Select a suitable cluster size based on the balance between available channels per cell and interference protection.

2. Interference reduction

Use sectorised antennas in high-interference regions to reduce unwanted co-channel interference.

3. Traffic management

Apply dynamic channel allocation to manage uneven traffic demand and reduce unnecessary channel blocking.

4. Scalable coverage

Introduce smaller cells in crowded locations instead of splitting every cell uniformly.

5. Capacity improvement

Monitor the number of active users and traffic load to identify cells requiring additional capacity.

6. Performance monitoring

Evaluate SIR, blocking probability, channel utilisation and handoff rate regularly.

**Performance Evaluation Metrics**

| Metric | Purpose |
|---|---|
| Signal-to-Interference Ratio | Measures signal quality relative to interference |
| Blocking Probability | Measures the probability that a new call cannot obtain a channel |
| Channel Utilisation | Measures the usage of allocated traffic channels |
| Traffic Capacity | Measures the amount of traffic supported by the network |
| Coverage Area | Measures the geographical region served by the network |
| Handoff Rate | Measures the frequency of mobile connections transferred between cells |

**DAY-4 DESIGN CHALLENGE**

**TASK 12 – Re-evaluate and Adapt the Design** **8 Marks**

**Changed Requirement**

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

**RETAIN → MODIFY → or REDESIGN**

your original solution.

**A. Impact Analysis – 2 Marks**

**Guiding Enquiry**

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

**Evidence Expected**

A clear analysis of how the surge affects the original design.

**B. Evaluation of the Original Design – 2 Marks**

**Guiding Enquiry**

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.

- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

**Evidence Expected**

A technically supported evaluation of the original design under the surge.

**C. Adaptation or Redesign – 2 Marks**

**Guiding Enquiry**

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

**Evidence Expected**

A design response with clear radio-planning reasoning and quantified effect.

**D. Evidence-Based Defence – 2 Marks**

**Guiding Enquiry**

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

**Evidence Expected**

A well-supported final decision integrating theory, simulation and the changed constraints.

**Calculation of Total Offered Traffic**

**The total offered traffic is calculated using:**

$$A = S \times a$$

Where:

- S = Number of subscribers.
- a = Traffic generated per subscriber.

Given:

$$S = 25000$$

$$a = 0.05\ Erlang$$

Substituting:

$$A = 25000 \times 0.05$$

$$\boxed{A = 1250\ Erlangs}$$

Therefore, the stadium network must support an offered traffic load of 1250 Erlangs.

**Calculation of Traffic Capacity per Cell**

The available channels per cell are:

$$C_{\text{cell}} = \frac{336}{7}$$

$$\boxed{C_{\text{cell}} = 48\ channels}$$

Using the Erlang-B model with 48 channels and a target blocking probability of 2%:

$$B(48, A) \approx 0.02$$

The approximate supported traffic is:

$$\boxed{A_{\text{cell}} \approx 38.39\ Erlangs}$$

The approximate number of subscribers supported by one cell is:

$$S_{\text{cell}} = \frac{38.39}{0.05}$$

$$\boxed{S_{\text{cell}} \approx 767\ subscribers}$$

**Calculation of Required Number of Cells**

The minimum number of cells required is:

$$N_{\text{cells}} = \frac{A_{\text{total}}}{A_{\text{cell}}}$$

$$N_{\text{cells}} = \frac{1250}{38.39}$$

$$N_{\text{cells}} = 32.56$$

Rounding up:

$$\boxed{N_{\text{cells}} = 33}$$

Therefore, at least 33 cells are theoretically required to support the estimated offered traffic under the assumed uniform traffic distribution.

**Recommended design with additional capacity reserve**

For a large public event, designing exactly for the expected traffic load provides little protection against unexpected congestion. Therefore, a 20% capacity reserve is proposed.

$$A_{\text{design}} = 1250 \times 1.20$$

$$A_{\text{design}} = 1500\ Erlangs$$

Required cells:

$$N_{\text{cells}} = \frac{1500}{38.39}$$

$$N_{\text{cells}} = 39.07$$

Rounding up:

$$\boxed{N_{\text{cells}} = 40}$$

**Hexagonal Cell Coverage Planning**

The area of a regular hexagonal cell is:

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2} R^2$$

The stadium area is 1 km².

For 40 cells, the approximate area allocated per cell is:

$$A_{\text{cell}} = \frac{1}{40}$$

$$A_{\text{cell}} = 0.025\ km^2$$

Therefore:

$$0.025 = \frac{3\sqrt{3}}{2}R^2$$

$$R^2 = \frac{0.025}{2.598}$$

$$R = \sqrt{0.00962}$$

$$\boxed{R \approx 0.098\ km}$$

Thus, the proposed cell radius is approximately 98 metres.

The frequency reuse distance is:

$$D = R\sqrt{3N}$$

$$D = 0.098\sqrt{21}$$

$$\boxed{D \approx 0.450\ km}$$

The smaller cell radius provides greater spatial reuse and helps support the high concentration of attendees.

**Frequency Allocation Strategy**

The proposed frequency reuse cluster size is 7.

$$C_{\text{cell}} = \frac{336}{7} = 48$$

| **Parameter** | **Proposed configuration** |
|---|---|
| Total available channels | 336 |
| Cluster size | 7 |
| Channels per cell | 48 |
| Number of cells | 40 |
| Approximate number of reuse clusters | 6 |
| Channel allocation | Repeated cluster-based allocation |

The 336 channels are divided into seven channel groups, with 48 channels assigned to each group. The groups are reused in other clusters according to the frequency reuse pattern.

**Interference Management**

The approximate SIR equation is:

$$SIR = \frac{(3N)^{n/2}}{i_0}$$

For the proposed design:

- $N = 7$
- $n = 3$
- $i_0 = 6$

Substituting:

$$SIR = \frac{(3 \times 7)^{3/2}}{6}$$

$$SIR = \frac{21^{1.5}}{6}$$

$$SIR = 16.04$$

Converting into decibels:

$$SIR_{dB} = 10 \log_{10}(16.04)$$

$$\boxed{SIR_{dB} \approx 12.05\ dB}$$

The calculated SIR is approximately 12.05 dB, which satisfies the proposed 12 dB target under the simplified six-interferer model.

Interference management techniques:

1. Apply frequency reuse with a cluster size of 7.
2. Use 120-degree sectorised antennas in high-density areas.
3. Apply suitable antenna orientation and transmission power control.
4. Monitor interference between adjacent cells.
5. Use dynamic channel allocation to manage temporary traffic variations.

**Comparison of Network Strategies**

| **Strategy** | **Advantages** | **Limitations** | **Application** |
|---|---|---|---|
| Fixed Frequency Reuse | Predictable channel allocation | Less flexible during traffic surges | Basic event coverage |
| Sectorisation | Reduces dominant interference | Requires additional antenna arrangements | Stadium seating areas |
| Dynamic Channel Allocation | Supports changing traffic demand | Requires central coordination | Crowded event locations |

| | | | |
|---|---|---|---|
| Small Cell Deployment | Improves local capacity | Requires additional infrastructure | Entrances, seating and food courts |

Recommended approach: Combine fixed frequency reuse, sectorisation and dynamic channel allocation. Deploy small cells in crowded regions such as spectator seating, entrances, food courts and ticket counters.

**Four-Hour Event Traffic Management Plan**

The four-hour duration is divided into operational stages to manage changes in attendee activity.

| **Event period** | **Expected network condition** | **Network management** |
|---|---|---|
| First hour | Arrival and ticket scanning | Increase capacity near entrances |
| Second hour | High usage during event activities | Activate all planned cells |
| Third hour | Heavy messaging, uploads and calls | Monitor congestion and allocate channels dynamically |
| Fourth hour | Event completion and attendee departure | Support outgoing traffic and maintain coverage near exits |

The traffic load may change throughout the event. Therefore, network monitoring and resource allocation should continue during all four hours.

**Final Proposed Network Configuration**

| **Design parameter** | **Final proposed value** |
|---|---|
| Stadium coverage area | 1 $km^2$ |
| Total attendees | 25,000 |
| Total offered traffic | 1,250 Erlangs |
| Design traffic with 20% reserve | 1,500 Erlangs |
| Minimum theoretical cells | 33 |
| Recommended cells | 40 |
| Cell radius | Approximately 0.098 km |
| Frequency reuse cluster size | 7 |
| Channels per cell | 48 |
| Approximate reuse distance | 0.450 km |
| Approximate capacity per cell | 38.39 Erlangs |
| Aggregate design capacity | 1,535.6 Erlangs |

| | |
|---|---|
| Target blocking probability | 2% |
| Approximate theoretical SIR | 12.05 dB |

The aggregate design capacity is:

$$A_{\text{total}} = 40 \times 38.39$$

$$\boxed{A_{\text{total}} = 1535.6\ Erlangs}$$

The capacity reserve is:

$$1535.6 - 1250 = 285.6\ Erlangs$$

This provides approximately 22.8% capacity above the expected offered traffic.

**MARK DISTRIBUTION**

| **Task** | **Marks** |
|---|---|
| Task 1 – Design Alternative Frequency-Planning Strategies | 5 |
| Task 2 – Identify the Governing Design Parameter | 2 |
| Task 3 – Derive the Capacity and Interference Functions Mathematically | 6 |
| Task 4 – Analyse Best, Average and Worst-Case Behaviour | 4 |
| Task 5 – Scaling and Sensitivity Analysis | 3 |
| Task 6 – Formulate the Recurrence for Repeated Cell Splitting | 3 |
| Task 7 – Solve the Recurrence and Interpret the Result | 4 |
| Task 8 – Implement and Test the Planning Strategies | 4 |
| Task 9 – Record and Present Experimental Performance | 3 |
| Task 10 – Compare Theory with Simulation | 4 |
| Task 11 – Recommend and Defend the Final Design | 4 |
| **Subtotal** | **42** |
| Task 12 – Day-4 Design Challenge | 8 |

| | |
|---|---|
| **TOTAL** | **50** |

**Submission Requirements (Digital)**

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

**Assessment Verification**

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

**Overall Assessment Expectation**

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

**Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate**

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.